Why Newton

Newton-Raphson linearizes nonlinear devices and iterates until the residual drops

The idea

- GOLDEN: 1 iteration with linear-only stamps; v2=3.0

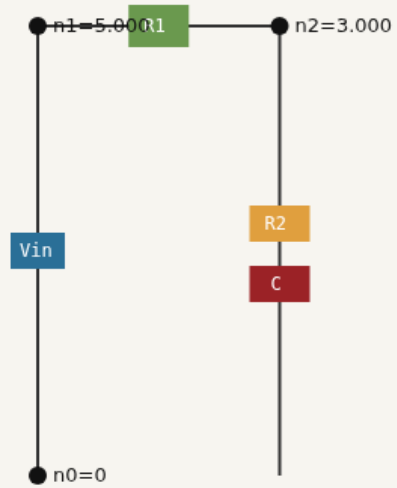
Guess

NEWTON–RAPHSON

Guess

Step 1 / 5 · guess · module03-01-newton-raphson

guess



Explanation

Start from linear OP.

- x0

```
linear
```

Guess

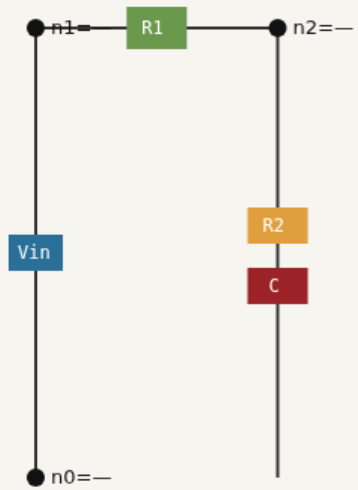
Jacobian

NEWTON–RAPHSON

Jacobian

Step 2 / 5 · jac · module03-01-newton-raphson

elements=5



Explanation

Restamp G each iteration.

- J

```
G(x)
```

Jacobian

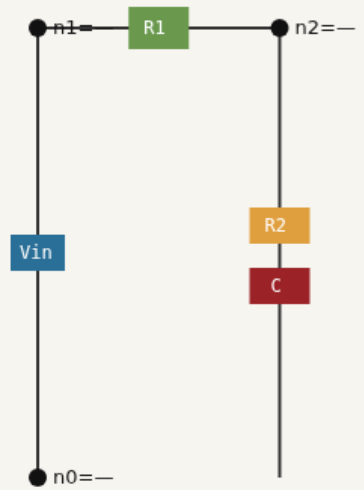
Update

NEWTON–RAPHSON

Update

Step 3 / 5 · upd · module03-01-newton-raphson

elements=5



Explanation

Solve and update x.

- dx

$$x \leftarrow x+dx$$

Update

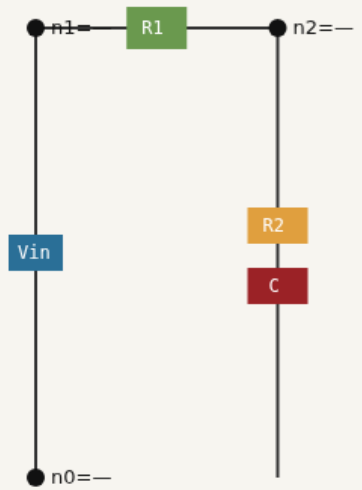
Residual

NEWTON–RAPHSON

Residual

Step 4 / 5 · res · module03-01-newton-raphson

elements=5



Explanation

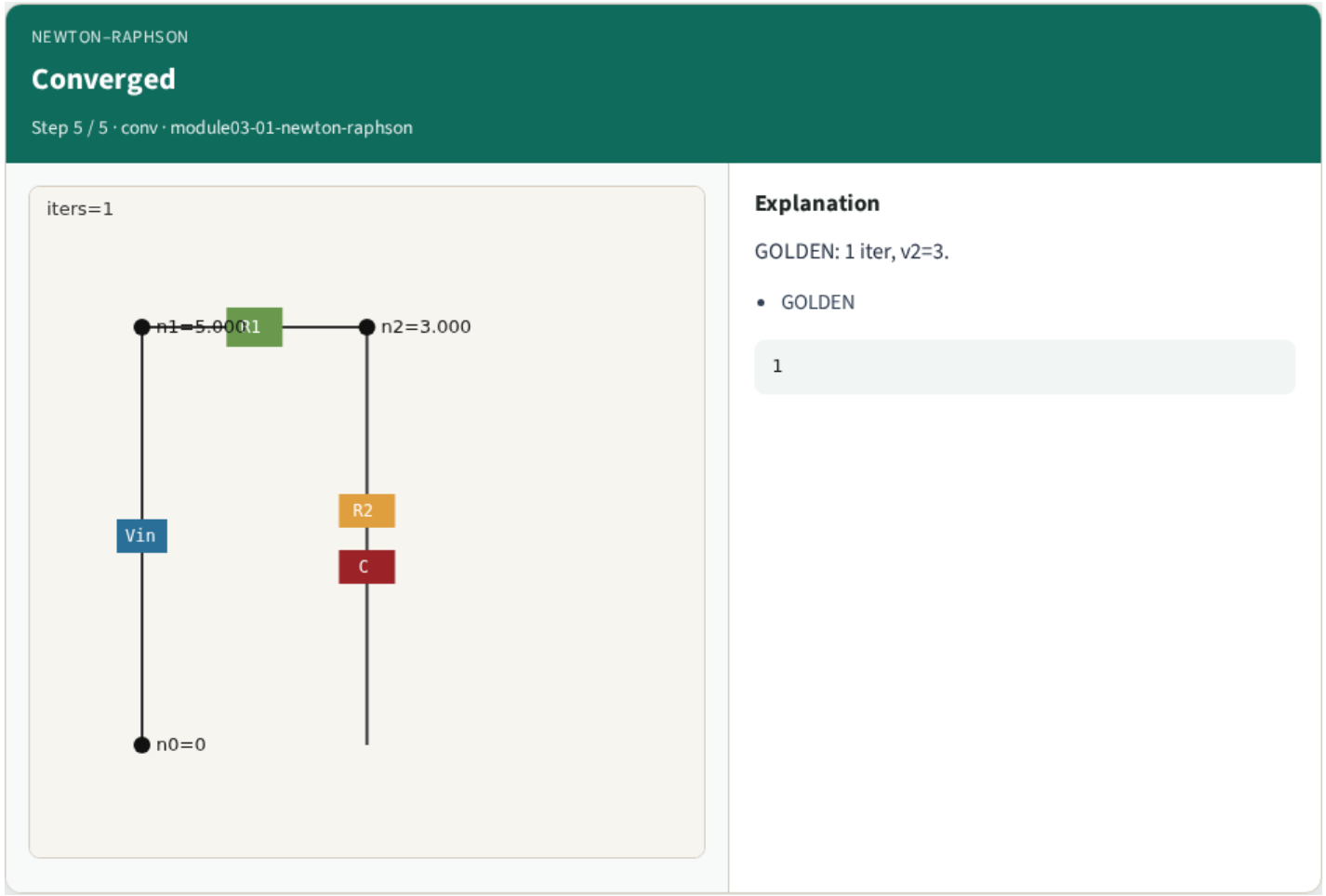
Stop when |dx| small.

- tol

```
1e-9
```

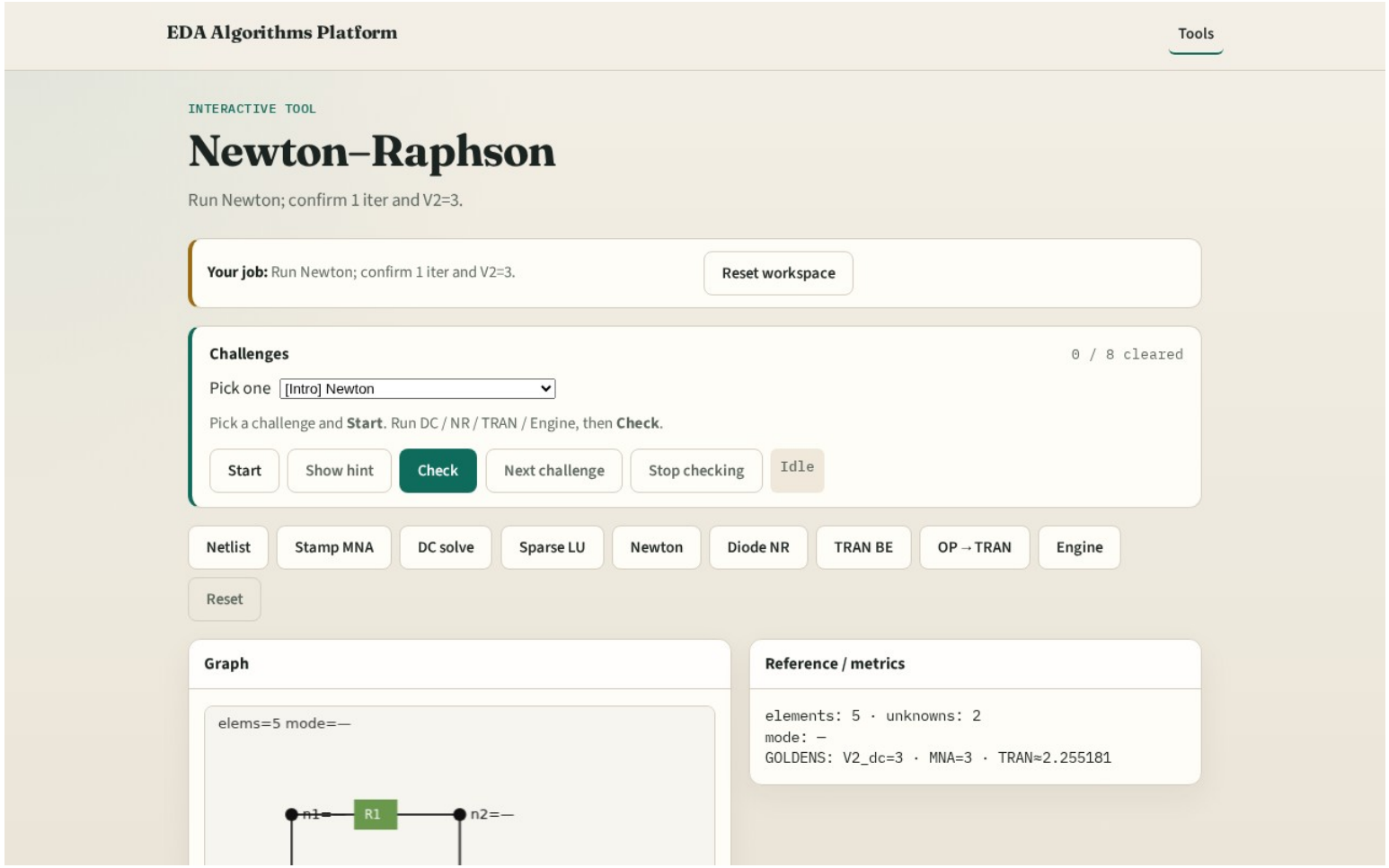
Residual

Converged



Converged

Browser lab track



Browser lab starter

Implement track

- Call newton_raphson_dc(include_diode=False); assert iters==1 and v2≈3.0

Pitfalls

- Enabling diode when the lesson is linear convergence
- Wrong tolerance units

Your turn

- Next: device callbacks for diodes